

N-Channel Logic Level Power MOSFET 60V, 11A, 107 mΩ

These N-Channel enhancement-mode power MOSFETs are manufactured using the latest manufacturing process technology. This process, which uses feature sizes approaching those of LSI circuits, gives optimum utilization of silicon, resulting in outstanding performance. They were designed for use in applications such as switching regulators, switching converters, motor drivers and relay drivers. These transistors can be operated directly from integrated circuits.

Formerly developmental type TA49158.

Ordering Information

PART NUMBER	PACKAGE	BRAND
RFD3055LE	TO-251AA	F3055L
RFD3055LESM9A	TO-252AA	F3055L

Features

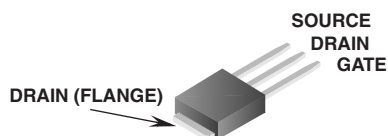
- 11A, 60V
- $r_{DS(ON)} = 0.107\Omega$
- Temperature Compensating PSPICE® Model
- Peak Current vs Pulse Width Curve
- UIS Rating Curve
- Related Literature
 - TB334 "Guidelines for Soldering Surface Mount Components to PC Boards"

Symbol



Packaging

JEDEC TO-251AA



JEDEC TO-252AA



RFD3055LE, RFD3055LESM

Absolute Maximum Ratings $T_C = 25^\circ\text{C}$, Unless Otherwise Specified

	RFD3055LE, RFD3055LESM9A	UNITS
Drain to Source Voltage (Note 1)	60	V
Drain to Gate Voltage ($R_{GS} = 20\text{k}\Omega$) (Note 1)	60	V
Gate to Source Voltage	± 16	V
Continuous Drain Current	11	A
Pulsed Drain Current (Note 3)	Refer to Peak Current Curve	
Single Pulse Avalanche Rating	Refer to UIS Curve	
Power Dissipation	38	W
Derate Above 25°C	0.25	W/ $^\circ\text{C}$
Operating and Storage Temperature	-55 to 175	$^\circ\text{C}$
Maximum Temperature for Soldering		
Leads at 0.063in (1.6mm) from Case for 10s.	300	$^\circ\text{C}$
Package Body for 10s, See Techbrief 334	260	$^\circ\text{C}$

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

NOTE:

1. $T_J = 25^\circ\text{C}$ to 150°C .

Electrical Specifications $T_C = 25^\circ\text{C}$, Unless Otherwise Specified

PARAMETER	SYMBOL	TEST CONDITIONS		MIN	TYP	MAX	UNITS
Drain to Source Breakdown Voltage	BV_{DSS}	$I_D = 250\mu A, V_{GS} = 0V$		60	-	-	V
Gate Threshold Voltage	$V_{GS(TH)}$	$V_{GS} = V_{DS}, I_D = 250\mu A$		1	-	3	V
Zero Gate Voltage Drain Current	I_{DSS}	$V_{DS} = 55V, V_{GS} = 0V$		-	-	1	μA
		$V_{DS} = 50V, V_{GS} = 0V, T_C = 150^{\circ}C$		-	-	250	μA
Gate to Source Leakage Current	I_{GSS}	$V_{GS} = \pm 16V$		-	-	± 100	nA
Drain to Source On Resistance (Note 2)	$r_{DS(ON)}$	$I_D = 8A, V_{GS} = 5V$ (Figure 11)		-	-	0.107	Ω
Turn-On Time	t_{ON}	$V_{DD} \approx 30V, I_D = 8A,$ $V_{GS} = 4.5V, R_{GS} = 32\Omega$ (Figures 10, 18, 19)		-	-	170	ns
Turn-On Delay Time	$t_{d(ON)}$			-	8	-	ns
Rise Time	t_r			-	105	-	ns
Turn-Off Delay Time	$t_{d(OFF)}$			-	22	-	ns
Fall Time	t_f			-	39	-	ns
Turn-Off Time	t_{OFF}			-	-	92	ns
Total Gate Charge	$Q_g(TOT)$	$V_{GS} = 0V$ to 10V	$V_{DD} = 30V, I_D = 8A,$ $I_{g(REF)} = 1.0mA$ (Figures 20, 21)	-	9.4	11.3	nC
Gate Charge at 5V	$Q_g(5)$	$V_{GS} = 0V$ to 5V		-	5.2	6.2	nC
Threshold Gate Charge	$Q_g(TH)$	$V_{GS} = 0V$ to 1V		-	0.36	0.43	nC
Input Capacitance	C_{ISS}	$V_{DS} = 25V, V_{GS} = 0V, f = 1MHz$ (Figure 14)		-	350	-	pF
Output Capacitance	C_{OSS}			-	105	-	pF
Reverse Transfer Capacitance	C_{RSS}			-	23	-	pF
Thermal Resistance Junction to Case	$R_{\theta JC}$			-	-	3.94	$^{\circ}C/W$
Thermal Resistance Junction to Ambient	$R_{\theta JA}$	TO-220AB		-	-	62	$^{\circ}C/W$
		TO-251AA, TO-252AA		-	-	100	$^{\circ}C/W$

Source to Drain Diode Specifications

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
Source to Drain Diode Voltage	V_{SD}	$I_{SD} = 8\text{A}$		-	1.25	V
Diode Reverse Recovery Time	t_{rr}	$I_{SD} = 8\text{A}$, $dI_{SD}/dt = 100\text{A}/\mu\text{s}$		-	66	ns

NOTES:

2. Pulse Test: Pulse Width $\leq 300\text{ms}$, Duty Cycle $\leq 2\%$.
3. Repetitive Rating: Pulse Width limited by max junction temperature. See Transient Thermal Impedance Curve (Figure 3) and Peak Current Capability Curve (Figure 5).

Typical Performance Curves Unless Otherwise Specified

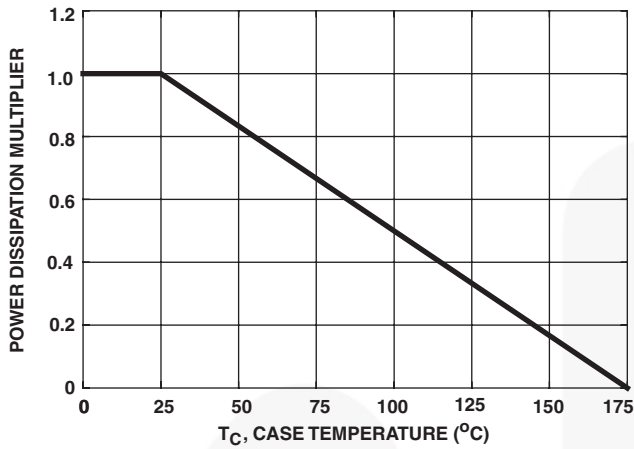


FIGURE 1. NORMALIZED POWER DISSIPATION vs CASE TEMPERATURE

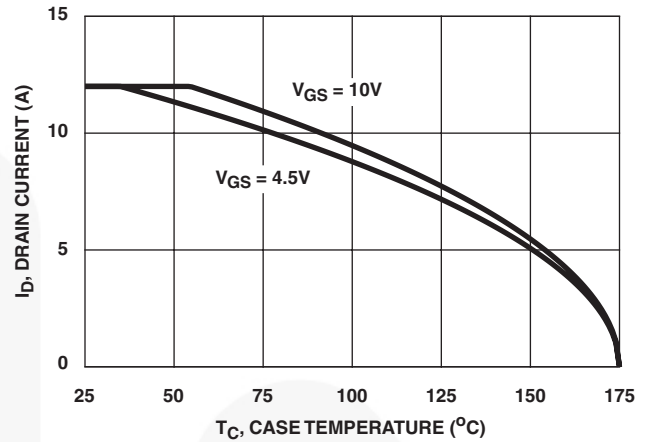


FIGURE 2. MAXIMUM CONTINUOUS DRAIN CURRENT vs CASE TEMPERATURE

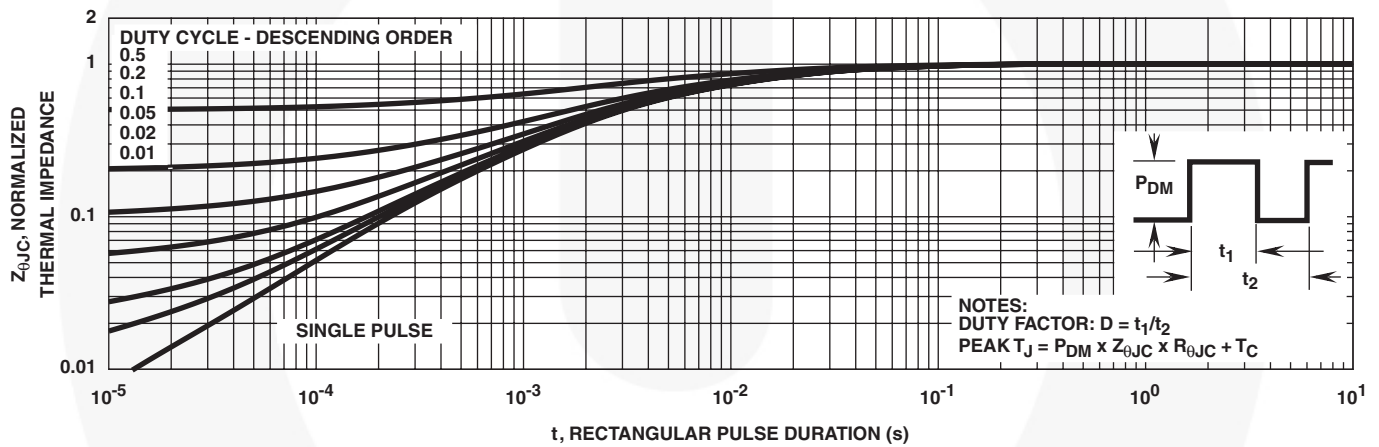


FIGURE 3. NORMALIZED TRANSIENT THERMAL IMPEDANCE

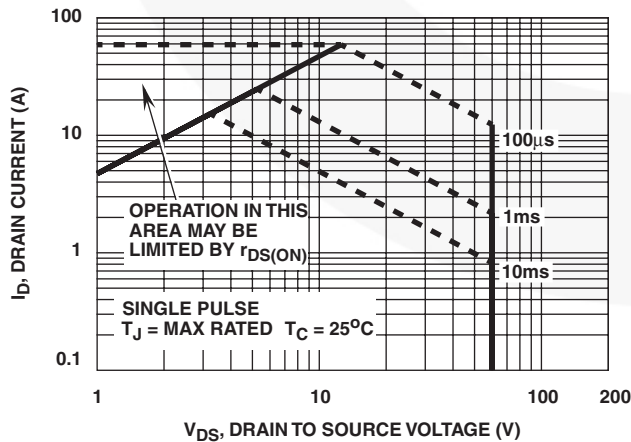


FIGURE 4. FORWARD BIAS SAFE OPERATING AREA

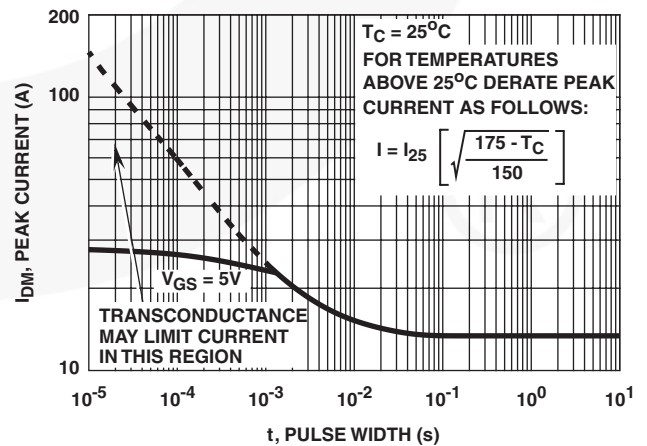
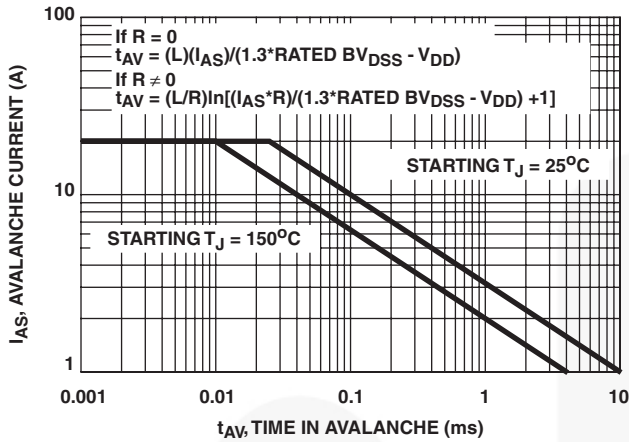


FIGURE 5. PEAK CURRENT CAPABILITY

Typical Performance Curves Unless Otherwise Specified (Continued)



NOTE: Refer to Fairchild Application Notes AN9321 and AN9322

FIGURE 6. UNCLAMPED INDUCTIVE SWITCHING

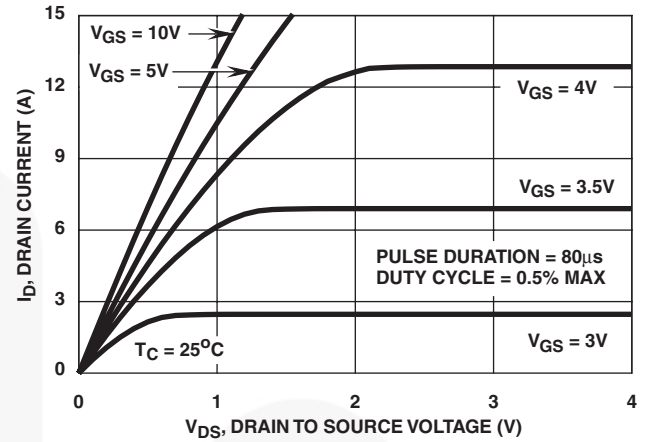


FIGURE 7. SATURATION CHARACTERISTICS

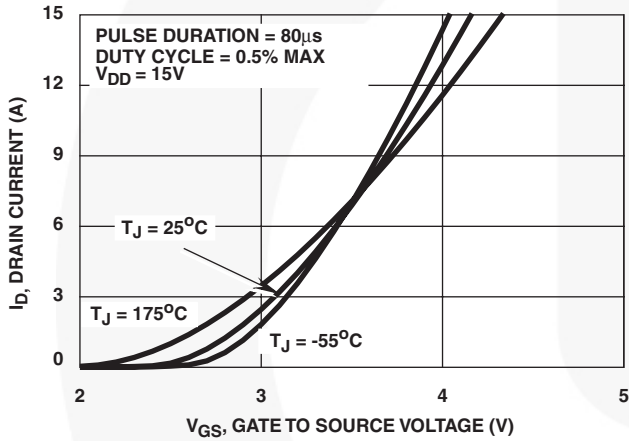


FIGURE 8. TRANSFER CHARACTERISTICS

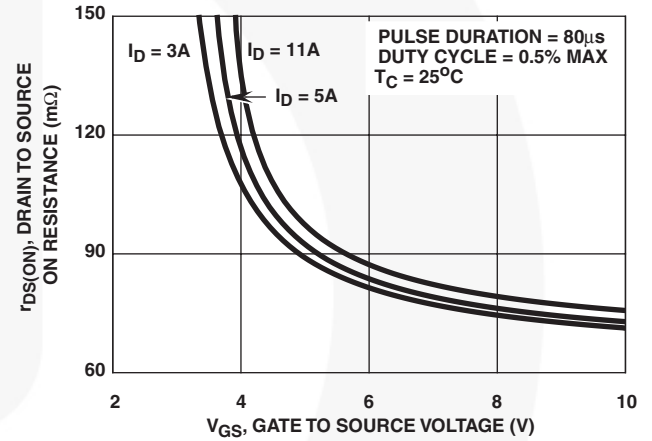


FIGURE 9. DRAIN TO SOURCE ON RESISTANCE vs GATE VOLTAGE AND DRAIN CURRENT

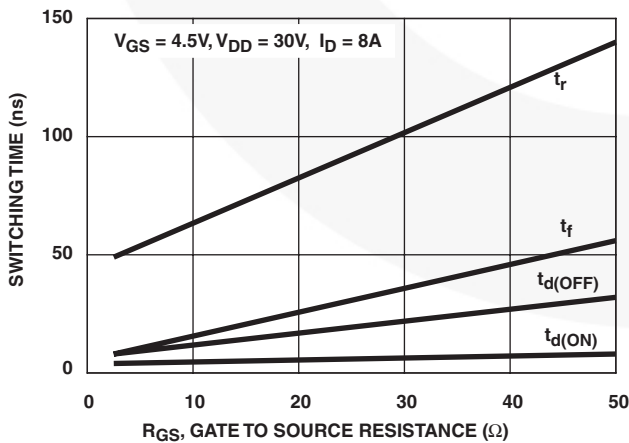


FIGURE 10. SWITCHING TIME vs GATE RESISTANCE

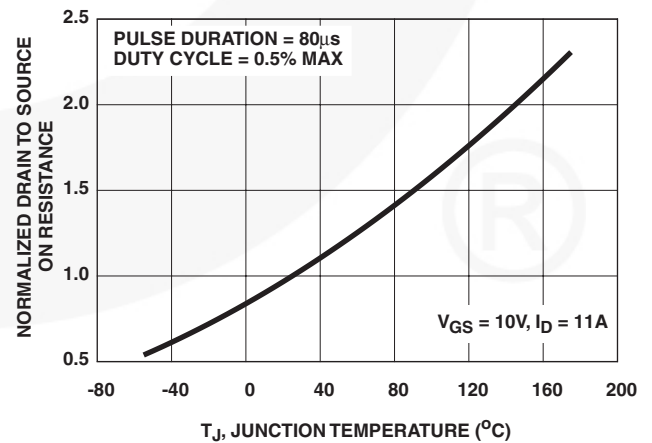


FIGURE 11. NORMALIZED DRAIN TO SOURCE ON RESISTANCE vs JUNCTION TEMPERATURE

Typical Performance Curves Unless Otherwise Specified (Continued)

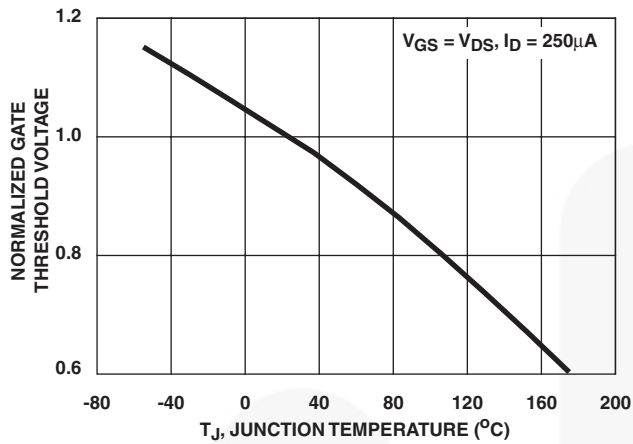


FIGURE 12. NORMALIZED GATE THRESHOLD VOLTAGE vs JUNCTION TEMPERATURE

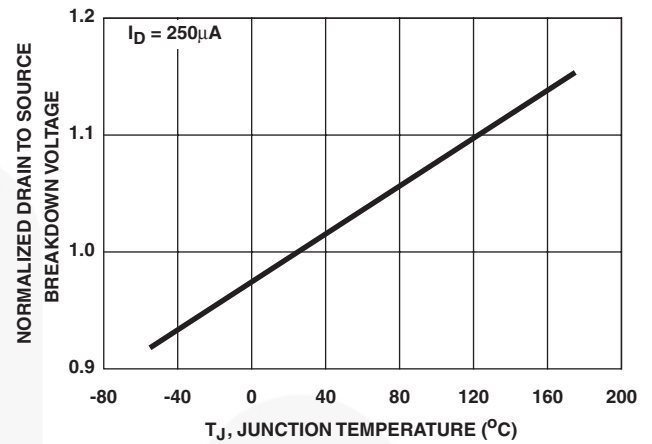


FIGURE 13. NORMALIZED DRAIN TO SOURCE BREAKDOWN VOLTAGE vs JUNCTION TEMPERATURE

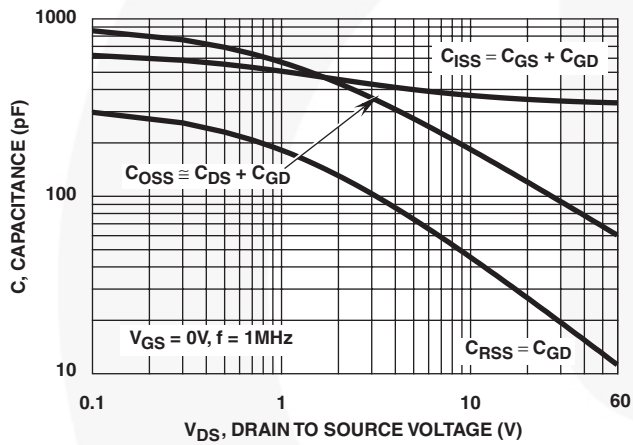
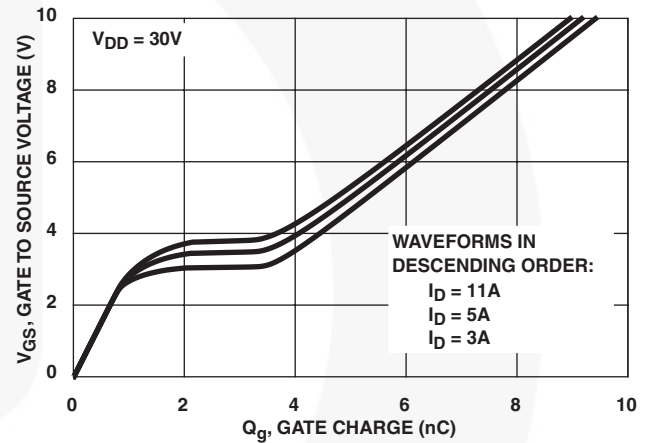


FIGURE 14. CAPACITANCE vs DRAIN TO SOURCE VOLTAGE



NOTE: Refer to Fairchild Application Notes AN7254 and AN7260.

FIGURE 15. NORMALIZED SWITCHING WAVEFORMS FOR CONSTANT GATE CURRENT

Test Circuits and Waveforms

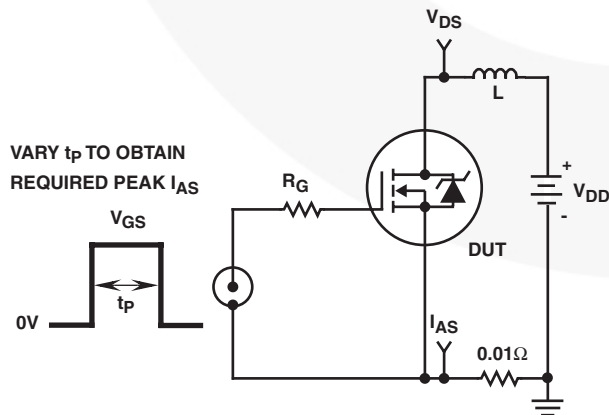


FIGURE 16. UNCLAMPED ENERGY TEST CIRCUIT

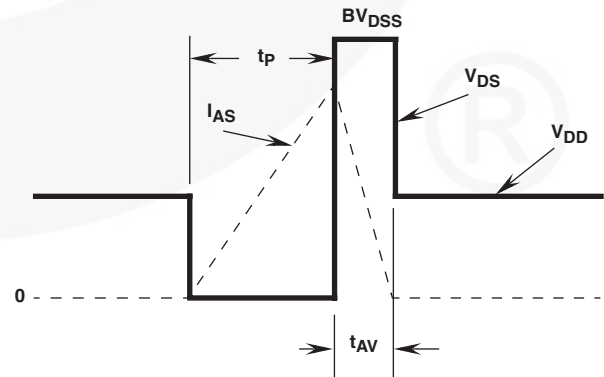


FIGURE 17. UNCLAMPED ENERGY WAVEFORMS

Test Circuits and Waveforms (Continued)

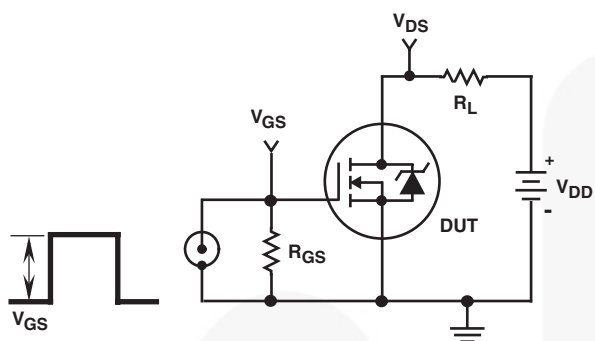


FIGURE 18. SWITCHING TEST CIRCUIT

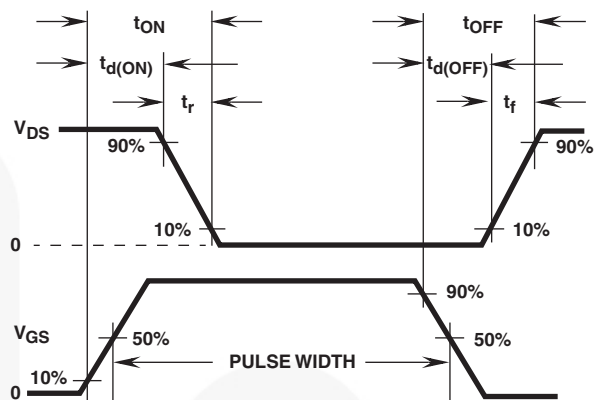


FIGURE 19. RESISTIVE SWITCHING WAVEFORMS

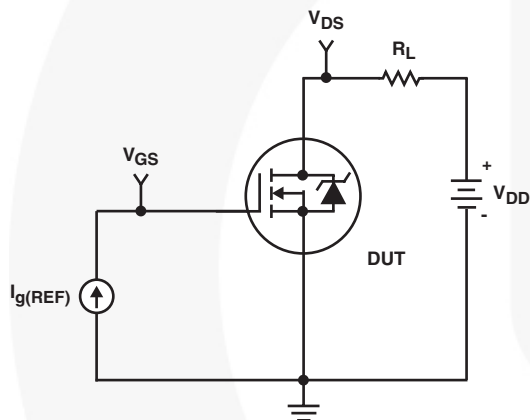


FIGURE 20. GATE CHARGE TEST CIRCUIT

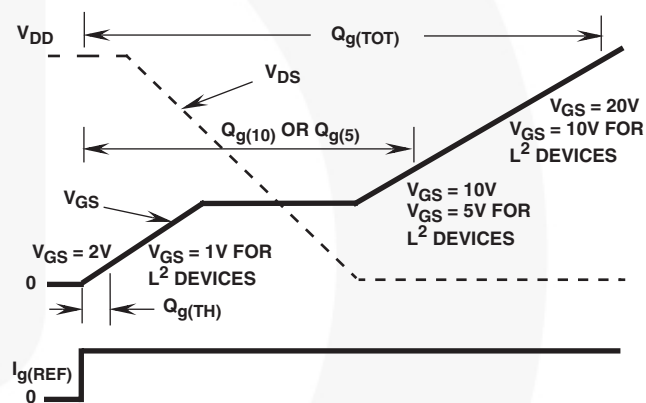


FIGURE 21. GATE CHARGE WAVEFORMS

PSpice Electrical Model

.SUBCKT RFD3055LE 2 1 3; rev 1/30/95

CA 12 8 3.9e-9
 CB 15 14 4.9e-9
 CIN 6 8 3.25e-10

DBODY 7 5 DBODYMOD
 DBREAK 5 11 DBREAKMOD
 DPLCAP 10 5 DPLCAPMOD

EBREAK 11 7 17 18 67.8
 EDS 14 8 5 8 1
 EGS 13 8 6 8 1
 ESG 6 10 6 8 1
 EVTHRES 6 21 19 8 1
 EVTEMP 20 6 18 22 1

IT 8 17 1

LDRAIN 2 5 1.0e-9
 LGATE 1 9 5.42e-9
 LSOURCE 3 7 2.57e-9

MMED 16 6 8 8 MMEDMOD
 MSTRO 16 6 8 8 MSTROMOD
 MWEAK 16 21 8 8 MWEAKMOD

RBREAK 17 18 RBREAKMOD 1
 RDRAIN 50 16 RDRAINMOD 3.7e-2
 RGATE 9 20 3.37
 RLDRAIN 2 5 10
 RLGATE 1 9 54.2
 RLSOURCE 3 7 25.7
 RSLC1 5 51 RSLCMOD 1e-6
 RSLC2 5 50 1e3
 RSOURCE 8 7 RSOURCEMOD 2.50e-2
 RVTHRES 22 8 RVTHRESMOD 1
 RVTEMP 18 19 RVTEMPMOD 1

S1A 6 12 13 8 S1AMOD
 S1B 13 12 13 8 S1BMOD
 S2A 6 15 14 13 S2AMOD
 S2B 13 15 14 13 S2BMOD

VBAT 22 19 DC 1

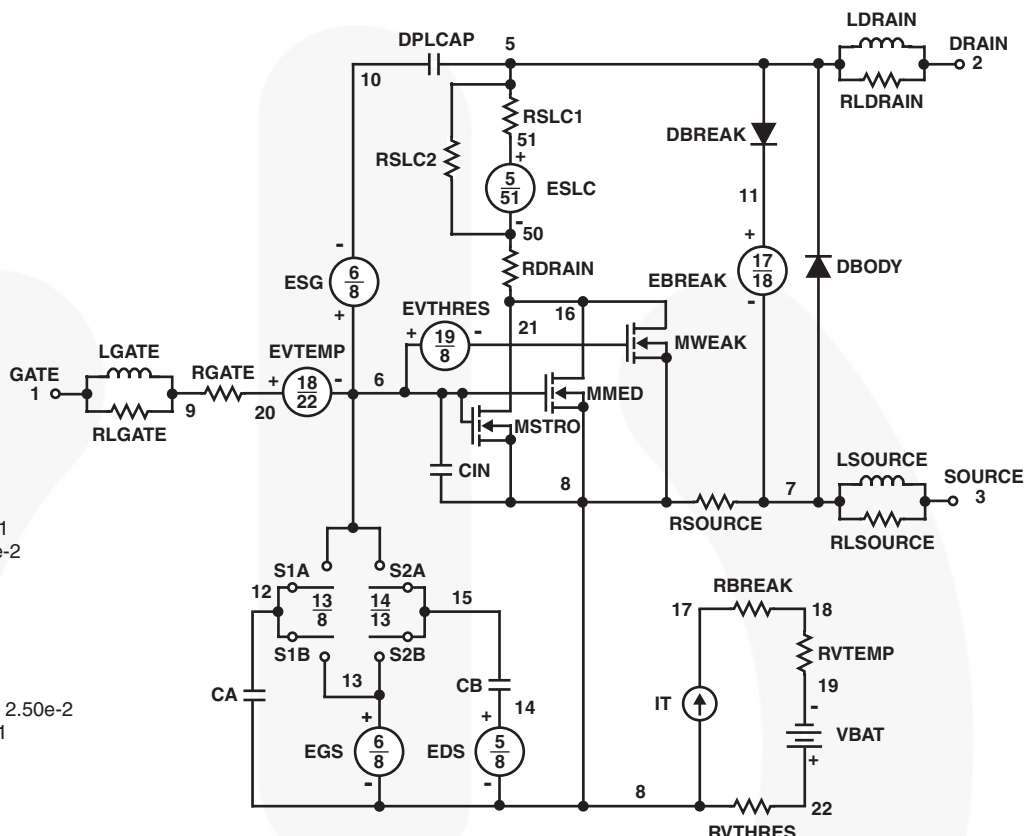
ESLC 51 50 VALUE={(V(5,51)/ABS(V(5,51)))*(PWR(V(5,51)/(1e-6*30),3))}

.MODEL DBODYMOD D (IS = 1.75e-13 RS = 1.75e-2 TRS1 = 1e-4 TRS2 = 5e-6 CJO = 5.9e-10 TT = 5.45e-8 N = 1.03 M = 0.6)
 .MODEL DBREAKMOD D (RS = 6.50e-1 TRS1 = 1.25e-4 TRS2 = 1.34e-6)
 .MODEL DPLCAPMOD D (CJO = 3.21e-10 IS = 1e-30 N = 10 M = 0.81)
 .MODEL MMEDMOD NMOS (VTO = 2.02 KP = .83 IS = 1e-30 N = 10 TOX = 1 L = 1u W = 1u RG = 3.37)
 .MODEL MSTROMOD NMOS (VTO = 2.39 KP = 14 IS = 1e-30 N = 10 TOX = 1 L = 1u W = 1u)
 .MODEL MWEAKMOD NMOS (VTO = 1.78 KP = 0.02 IS = 1e-30 N = 10 TOX = 1 L = 1u W = 1u RG = 33.7 RS = 0.1)
 .MODEL RBREAKMOD RES (TC1 = 1.06e-3 TC2 = 0)
 .MODEL RDRAINMOD RES (TC1 = 1.23e-2 TC2 = 2.58e-5)
 .MODEL RSLCMOD RES (TC1 = 0 TC2 = 0)
 .MODEL RSOURCEMOD RES (TC1 = 1e-3 TC2 = 0)
 .MODEL RVTHRESMOD RES (TC1 = -2.19e-3 TC2 = -4.97e-6)
 .MODEL RVTEMPMOD RES (TC1 = -1.6e-3 TC2 = 1e-7)

.MODEL S1AMOD VSWITCH (RON = 1e-5 ROFF = 0.1 VON = -4 VOFF = -2.5)
 .MODEL S1BMOD VSWITCH (RON = 1e-5 ROFF = 0.1 VON = -2.5 VOFF = -4)
 .MODEL S2AMOD VSWITCH (RON = 1e-5 ROFF = 0.1 VON = -0.5 VOFF = 0)
 .MODEL S2BMOD VSWITCH (RON = 1e-5 ROFF = 0.1 VON = 0 VOFF = -0.5)

.ENDS

For further discussion of the PSpice model, consult **A New PSpice Sub-Circuit for the Power MOSFET Featuring Global Temperature Options**; IEEE Power Electronics Specialist Conference Records, 1991, written by William J. Hepp and C. Frank Wheatley.





TRADEMARKS

The following includes registered and unregistered trademarks and service marks, owned by Fairchild Semiconductor and/or its global subsidiaries, and is not intended to be an exhaustive list of all such trademarks.

AccuPower™	F-PFS™	PowerTrench®	Sync-Lock™
AX-CAP®*	FRFET®	PowerXS™	SYSTEM®*
BitSiC™	Global Power ResourceSM	Programmable Active Droop™	TinyBoost®
Build it Now™	GreenBridge™	QFET®	TinyBuck®
CorePLUS™	Green FPS™	QS™	TinyCalc™
CorePOWER™	Green FPS™ e-Series™	Quiet Series™	TinyLogic®
CROSSVOLT™	Gmax™	RapidConfigure™	TINYOPTO™
CTL™	GTO™	Saving our world, 1mW/W/kW at a time™	TinyPower™
Current Transfer Logic™	IntelliMAX™	SignalWise™	TinyPWM™
DEUXPEED®	ISOPLANAR™	SmartMax™	TinyWire™
Dual Cool™	Marking Small Speakers Sound Louder and Better™	SMART START™	TranSiC™
EcoSPARK®	MegaBuck™	Solutions for Your Success™	TriFault Detect™
EfficientMax™	MICROCOUPLER™	SPM®	TRUECURRENT®*
ESBC™	MicroFET™	STEALTH™	μSerDes™
	MicroPak™	SuperFET®	UHC®
Fairchild®	MicroPak2™	SuperSOT™-3	Ultra FRFET™
Fairchild Semiconductor®	MillerDrive™	SuperSOT™-6	UniFET™
FACT Quiet Series™	MotionMax™	SuperSOT™-8	VCX™
FACT®	mWSaver®	SupreMOS®	VisualMax™
FAST®	OptoHiT™	SyncFET™	VoltagePlus™
FastvCore™	OPTOLOGIC®		XS™
FETBench™	OPTOPLANAR®		
FPS™			

*Trademarks of System General Corporation, used under license by Fairchild Semiconductor.

DISCLAIMER

FAIRCHILD SEMICONDUCTOR RESERVES THE RIGHT TO MAKE CHANGES WITHOUT FURTHER NOTICE TO ANY PRODUCTS HEREIN TO IMPROVE RELIABILITY, FUNCTION, OR DESIGN. FAIRCHILD DOES NOT ASSUME ANY LIABILITY ARISING OUT OF THE APPLICATION OR USE OF ANY PRODUCT OR CIRCUIT DESCRIBED HEREIN; NEITHER DOES IT CONVEY ANY LICENSE UNDER ITS PATENT RIGHTS, NOR THE RIGHTS OF OTHERS. THESE SPECIFICATIONS DO NOT EXPAND THE TERMS OF FAIRCHILD'S WORLDWIDE TERMS AND CONDITIONS, SPECIFICALLY THE WARRANTY THEREIN, WHICH COVERS THESE PRODUCTS.

LIFE SUPPORT POLICY

FAIRCHILD'S PRODUCTS ARE NOT AUTHORIZED FOR USE AS CRITICAL COMPONENTS IN LIFE SUPPORT DEVICES OR SYSTEMS WITHOUT THE EXPRESS WRITTEN APPROVAL OF FAIRCHILD SEMICONDUCTOR CORPORATION.

As used here in:

1. Life support devices or systems are devices or systems which, (a) are intended for surgical implant into the body or (b) support or sustain life, and (c) whose failure to perform when properly used in accordance with instructions for use provided in the labeling, can be reasonably expected to result in a significant injury of the user.
2. A critical component in any component of a life support, device, or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

ANTI-COUNTERFEITING POLICY

Fairchild Semiconductor Corporation's Anti-Counterfeiting Policy. Fairchild's Anti-Counterfeiting Policy is also stated on our external website, www.Fairchildsemi.com, under Sales Support.

Counterfeiting of semiconductor parts is a growing problem in the industry. All manufacturers of semiconductor products are experiencing counterfeiting of their parts. Customers who inadvertently purchase counterfeit parts experience many problems such as loss of brand reputation, substandard performance, failed application, and increased cost of production and manufacturing delays. Fairchild is taking strong measures to protect ourselves and our customers from the proliferation of counterfeit parts. Fairchild strongly encourages customers to purchase Fairchild parts either directly from Fairchild or from Authorized Fairchild Distributors who are listed by country on our web page cited above. Products customers buy either from Fairchild directly or from Authorized Fairchild Distributors are genuine parts, have full traceability, meet Fairchild's quality standards for handling and storage and provide access to Fairchild's full range of up-to-date technical and product information. Fairchild and our Authorized Distributors will stand behind all warranties and will appropriately address and warranty issues that may arise. Fairchild will not provide any warranty coverage or other assistance for parts bought from Unauthorized Sources. Fairchild is committed to combat this global problem and encourage our customers to do their part in stopping this practice by buying direct or from authorized distributors.

PRODUCT STATUS DEFINITIONS

Definition of Terms

Datasheet Identification	Product Status	Definition
Advance Information	Formative / In Design	Datasheet contains the design specifications for product development. Specifications may change in any manner without notice.
Preliminary	First Production	Datasheet contains preliminary data; supplementary data will be published at a later date. Fairchild Semiconductor reserves the right to make changes at any time without notice to improve design.
No Identification Needed	Full Production	Datasheet contains final specifications. Fairchild Semiconductor reserves the right to make changes at any time without notice to improve the design.
Obsolete	Not In Production	Datasheet contains specifications on a product that is discontinued by Fairchild Semiconductor. The datasheet is for reference information only.

Rev. I66

Mouser Electronics

Authorized Distributor

Click to View Pricing, Inventory, Delivery & Lifecycle Information:

[ON Semiconductor:](#)

[RFD3055LE_R4821](#) [RFD3055LE_Q](#)